

BHARATH SIMHA REDDY MADATALA

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Education

Indian Institute of Information Technology Design and Manufacturing, Jabalpur

B.Tech in Computer Science and Engineering

2023 – Present

Coursework: Operating Systems, Data Structures and Algorithms, Design and Analysis of Algorithms, Computer Networks, DBMS, OOPs in Java, Programming in C

Projects

Crop and Fertilizer Recommendation System

- Built a precision agriculture recommendation system to predict optimal crops from soil and weather parameters and suggest fertilizers based on soil, crop, and nutrient profiles
- Processed and analyzed **10,000+ agricultural records**, achieving **96% crop prediction accuracy** with **RandomForestClassifier** and **93% fertilizer prediction accuracy** with **GradientBoostingClassifier**
- Implemented end-to-end ML pipeline including data preprocessing, feature engineering, model evaluation, and performance optimization

AI-Powered Diet Recommender using Food Image Recognition

- Developed an end-to-end food recognition and nutrition analysis pipeline using **EfficientNetV2-S**, achieving up to **88% classification accuracy** on the **Food-101 dataset**
- Enabled calorie estimation with a **10–15% error margin**, improving dietary tracking accuracy
- Utilized **PyTorch**, **FastAPI**, **NumPy**, and **Scikit-learn** for model training, backend development, and performance optimization

Technical Skills

- **Languages & Scripting:** Python, C++, Java, SQL
- **Frameworks & Libraries:** Scikit-learn, Pandas, NumPy, PySpark, Image Classification
- **Tools & Platforms:** Git
- **Databases:** MongoDB, MySQL

Positions of Responsibility

Student Senate Member

Jan 2025 - Present

- Elected with the highest number of votes, representing **500+ students** and influencing campus policies
- Led multiple initiatives and coordinated with administration, improving issue resolution efficiency by **30%**

Core Member, Shutterbox Photography Club

- Organized **10+ workshops and events** including photo walks and exhibitions with active student participation

Awards and Achievements

- Solved 700+ problems on LeetCode across core data structures and algorithms
- Achieved **Knight rating on LeetCode**, demonstrating strong problem-solving and competitive programming skills

Extracurricular

- **DSLR Photography:** Skilled in composition, lighting, and post-processing techniques